Three-Pathway Pharmacokinetic–Pharmacodynamic Analysis of Levodopa in Parkinson's Disease: DaT-SPECT-Calibrated Neurodegeneration Moderates Treatment Benefit via an $N(t) \times \mathrm{LEDD}$ Interaction

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Study Highlights

What is the current knowledge on the topic? Published Parkinson's disease (PD) PK/PD models treat neurodegeneration as a time-invariant covariate. No prior work tests whether per-patient imaging-calibrated dopaminergic neuron fraction $N(t)/N_0$ [7] moderates levodopa benefit.

What question did this study address? Does DaT-SPECT-calibrated $N(t)/N_0$ predict the ON–OFF UPDRS-III gap (treatment benefit), OFF-state motor trajectory, or wearing-off timing in 1,065 PPMI patients?

What does this study add to our knowledge? Across 3,178 paired ON–OFF visits in 772 patients, the $N(t) \times \mathrm{LEDD}$ interaction is directionally positive ($\Delta\mathrm{AIC} = -72$; Bayesian $P(\beta_3 > 0) = 0.991$; frequentist $p = 0.011$): each additional unit of LEDD enlarges the motor gap more in patients with more surviving neurons. OFF-state trajectory is better explained by elapsed time ($\Delta\mathrm{AIC} = +803$, informative negative) and wearing-off is PK-driven (C-index $= 0.515$, informative negative). The sub-EC_{50} linear regime is confirmed (free Hill $h = 0.13$).

How might this change clinical pharmacology or translational science? $N(t)/N_0$ is a deployment-relevant biomarker for attenuated medication response in advanced-stage PD. The Path B coefficients reported here are operationalised by a companion bidirectional-ready mechanistic twin [5] via Sequential Importance Resampling posterior updates, and surfaced at point-of-care by a companion clinical decision-support pipeline [6] for 1,065 of 1,900 PPMI patients.

Abstract

Background. No published model couples per-patient imaging-calibrated neurodegeneration $N(t)$ [7] to longitudinal levodopa response in Parkinson's disease. We test whether DaT-SPECT-derived neuron fraction N_{frac} interacts with levodopa-equivalent daily dose (LEDD) to predict motor benefit, OFF trajectory, and wearing-off.

Methods. In PPMI (n=1,065 patients with Bayesian $N(t)$, 22,270 UPDRS-III assessments, 1,674 with LEDD), we performed a three-pathway analysis: (A) OFF-state UPDRS-III vs. N_{frac}; (B) ON–OFF gap vs. $N_{\text{frac}} \times$ LEDD; (C) wearing-off onset vs. neurodegeneration rate. Methods: mixed-effects models, Hill dose–response, Cox proportional hazards, formal identifiability analysis.

Results. Path B was the headline finding: the $N_{\text{frac}} \times$ LEDD interaction predicts the ON–OFF gap (n=3,178 paired visits, 772 patients; ΔAIC=-72; $\beta(N_{\text{frac}})=-11.62$, $p<10^{-37}$; conditional R^2=0.491). Each 10% decrease in N_{frac} enlarged the ON–OFF gap by 1.16 UPDRS-III points. The Hill model failed (free h=0.13), confirming a sub-EC$_{50}$ linear regime. Paths A and C were informative negatives: N_{frac} did not outperform elapsed time for OFF-state trajectory (ΔAIC=+803), and neurodegeneration rate did not predict wearing-off ($\rho=-0.050$, p=0.43; C-index=0.515).

Discussion. N_{frac} moderates levodopa benefit via an interaction, not a main effect. Wearing-off is PK-driven. These findings support $N(t)$-informed LEDD optimisation and motivate per-patient mechanistic digital twins [5] for treatment planning.

Keywords: DaT-SPECT; digital twin; dopaminergic neurodegeneration; Hill model; levodopa; mixed-effects model; NSD-ISS; Parkinson's disease; pharmacodynamics; PPMI; UPDRS-III; wearing-off

Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by the loss of dopaminergic neurons in the substantia nigra pars compacta (SNc). The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS) provides a biologically grounded framework for classifying disease severity based on synucleinopathy and dopaminergic deficit biomarkers [23]. DaT-SPECT imaging quantifies striatal dopamine transporter density, serving as an in vivo proxy for surviving dopaminergic neuron fraction [21].

Levodopa remains the gold-standard treatment for PD motor symptoms. However, as disease progresses and dopaminergic neurons are lost, treatment benefit diminishes and motor complications such as wearing-off emerge. The LEAP trial [26] and its 5-year follow-up [12] established that levodopa has no disease-modifying effect on neurodegeneration—confirming that LEDD reflects treatment for symptoms, not a cause of neuron death. Post-mortem studies further support this conclusion: Parkkinen *et al.* [18] found no association between levodopa exposure and nigral neuron counts, and animal models show that L-dopa does not affect DAT density [11]. This evidence permits us to model LEDD as a proxy for unmeasured symptom severity, not a causal mechanism of neurodegeneration.

Several pharmacometric models have addressed PD progression and levodopa response. Holford *et al.* [14] developed empirical nonlinear mixed-effects (NLME) models linking time to levodopa pharmacodynamics without imaging biomarkers. The K-PD framework of Jacqmin *et al.* [15] models drug effects in the absence of measured concentrations. More recently, Gupta *et al.* [13] proposed a biomarker-directed clinical endpoint model linking DaT-SPECT striatal binding ratio (SBR) to MDS-UPDRS via item response theory (IRT) in 419 PPMI patients, but without a medication covariate. Chae *et al.* [3] predicted longitudinal changes in LEDD requirements using IRT-assessed UPDRS. Véronneau-Veilleux *et al.* [25] developed an integrative model coupling levodopa pharmacokinetics to basal ganglia neurotransmission, but used generic—not per-patient, imaging-calibrated—neurodegeneration trajectories. Ursino *et al.* [24] modeled levodopa motor response with parameter estimation, and Simon *et al.* [22] combined pharmacokinetic/pharmacodynamic modeling of levodopa motor response and dyskinesia. Angiolelli *et al.* [1] proposed a Virtual Parkinsonian Patient framework, and Rukavina *et al.* [20] showed that DaT uptake predicts subsequent

LEDD requirements.

A critical gap remains: no published model couples *per-patient, imaging-calibrated* dopaminergic neurodegeneration trajectories $N(t)$ to levodopa treatment response. We address this gap by linking Bayesian posterior $N(t)$ trajectories—derived from serial DaT-SPECT scans in 1,065 PPMI patients—to longitudinal UPDRS-III outcomes and LEDD through a three-pathway mechanistic analysis framework.

Our approach is motivated by the following mechanistic reasoning: levodopa is converted to dopamine by aromatic L-amino acid decarboxylase (AADC) in surviving nigrostriatal neurons. As neurons die, the capacity for this conversion diminishes, and treatment benefit should decline proportionally. This predicts a specific interaction effect: the motor benefit of a given LEDD should depend on N_{frac}. Rather than attempting to fit a full PK/PD model—which would require plasma drug levels not available in PPMI—we adopt a PD-only framework with patient-specific neurodegeneration calibrated from DaT-SPECT Bayesian posteriors, using LEDD as a dose proxy. We emphasise at the outset that this is not a population-PK-instantiated analysis: no plasma levodopa compartments, absorption kinetics, or clearance parameters are fit; LEDD serves as the dosing summary and the imaging-calibrated $N(t)/N_0$ anchors individual disease state.

We test five hypotheses through three analytical pathways:

H1: The $N_{\text{frac}} \times$ LEDD interaction predicts the ON–OFF gap better than either component alone.

H2: N_{frac} is the primary moderator of treatment benefit (negative coefficient in mixed-effects model).

H3: $N(t)$ does *not* predict wearing-off timing (informative negative).

H4: $N(t)$ does *not* outperform simple elapsed time for OFF-state UPDRS-III prediction (informative negative).

H5: PPMI patients operate in the sub-EC_{50} linear regime of the dose–response curve.

We report two positive findings (H1, H2), two pre-registered informative negatives (H3, H4), and one mechanistic insight (H5), demonstrating that honest reporting of negative results advances mechanistic understanding as much as positive findings.

Table 1: Cohort description and data sources.

Data source	Patients	Observations	Notes
UPDRS-III (OFF)	5,014	22,270	All available assessments
UPDRS-III (ON)	—	9,472	Same patients, ON-state
LEDD	1,674	8,970	Concomitant Medication Log
Part IV (wearing-off)	1,610	10,688	NP4OFF items
$N(t)$ posteriors	1,065	—	Phase 2 IS-weighted
NSD-ISS staging	1,900	16,699	Longitudinal staging
Intersection datasets:			
Path A (OFF + $N(t)$)	988	4,772	OFF-state motor trajectory
Path B (ON–OFF + $N(t)$ + LEDD)	772	3,178	Treatment benefit
Path C (Part IV + $N(t)$)	276	—	Wearing-off timing

Methods

Study Population

Data were obtained from the Parkinson's Progression Markers Initiative (PPMI), accessed via the AMP-PD platform (data freeze: April 12, 2026). The PPMI is a multicenter longitudinal observational study enrolling participants with early, untreated PD and healthy controls [23]. Our analysis draws on three overlapping patient populations (Table 1):

(i) **UPDRS-III cohort:** 5,014 patients with 22,270 OFF-state UPDRS-III assessments (37,399 total Part III records; 9,472 ON-state assessments).

(ii) **LEDD cohort:** 1,674 patients with 8,970 medication records from the Concomitant Medication Log (613 non-numeric LEDD entries excluded).

(iii) **Posterior cohort:** 1,065 patients with Bayesian posterior $N(t)$ trajectories from serial DaT-SPECT (Phase 2 importance-sampling weighted posteriors; 304 Wave A patients with $\geq$4 scans, 761 Wave B patients with 2–3 scans).

Data Sources

Levodopa Equivalent Daily Dose (LEDD)

LEDD values were extracted from the PPMI Concomitant Medication Log (downloaded April 12, 2026; 9,583 rows, 1,678 patients). Records with non-numeric LEDD entries (e.g., "LD $\times$ 0.33" for entacapone multipliers, n=613) were excluded. LEDD was summed across all active medications per visit. LEDD conversion factors follow the updated systematic review by Jost *et al.* [16]. For regression models, LEDD was scaled by dividing by 500 to improve numerical conditioning.

UPDRS-III Motor Examination

MDS-UPDRS Part III total scores (NP3TOT) were obtained from 37,399 records (5,014 patients). Scores were stratified by PDSTATE into OFF-state (n=22,270; includes records with missing PDSTATE, assumed OFF per PPMI protocol for early-stage patients) and ON-state (n=9,472). The ON–OFF gap was computed as Gap = $\text{UPDRS-III}_{\text{OFF}} - \text{UPDRS-III}_{\text{ON}}$ for visits with both assessments on the same date (n=4,203 paired visits, 1,220 patients). Mean gap was 9.42 ± 8.15 UPDRS-III points (median 8.0; range -40 to 90). Negative gaps (3.6% of pairs) occur when ON-state dyskinesia inflates the ON score.

Wearing-Off (MDS-UPDRS Part IV)

Motor complications were assessed via MDS-UPDRS Part IV (10,688 records, 1,610 patients). We defined wearing-off onset as the first visit with NP4OFF ≥ 1 (primary threshold) or NP4OFF ≥ 2 (sensitivity analysis). Time-to-event was computed from the first visit with PDMEDYN=1 (on PD medication) to the first visit meeting the threshold.

Per-Patient Neurodegeneration Trajectories $N(t)/N_0$

Patient-specific neuron fraction trajectories were derived from our Phase 2 mechanistic twin pipeline. Briefly, a coupled ODE model of α-synuclein aggregation and dopaminergic neuron death was calibrated per patient using serial DaT-SPECT SBR measurements (caudate mean) via importance-sampling (IS) weighted Bayesian inference [10]. The IS posterior was computed over 5,000 resamples per patient. From the posterior median neurodegeneration rate (percent loss per year), the neuron fraction at each visit was computed as:

$$\frac{N(t)}{N_0} = \left(1 - \frac{r}{100}\right)^t \tag{1}$$

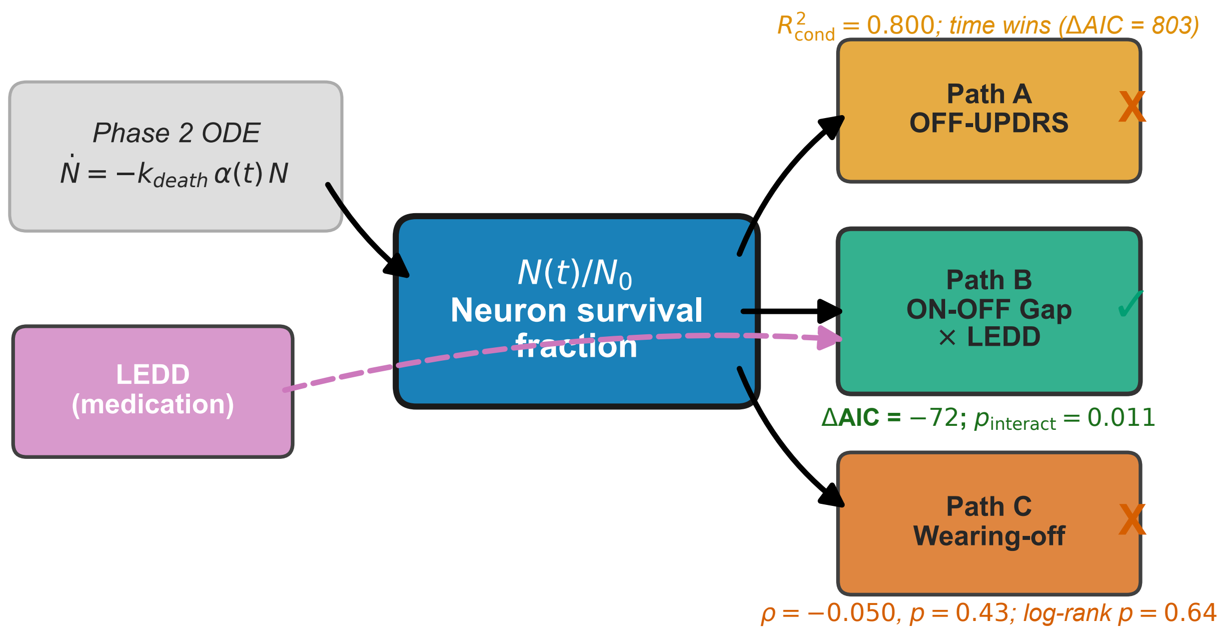

Figure 1: Three-pathway conceptual framework linking DaT-SPECT-calibrated neurodegeneration trajectories N_{frac} to levodopa treatment response. Path A tests whether N_{frac} predicts OFF-state motor trajectory; Path B tests whether the $N_{frac} \times$ LEDD interaction predicts the ON–OFF gap (treatment benefit); Path C tests whether neurodegeneration rate predicts wearing-off timing.

where r is the posterior median percent loss per year and t is time from baseline in years. The combined cohort (n=1,065: 304 Wave A + 761 Wave B) had a population median r=1.59%/yr (mean 4.79%/yr), consistent with the canonical 2–5%/yr range from clinicopathological studies [10].

Three-Pathway Analysis Framework

We decompose the levodopa–neurodegeneration coupling into three testable pathways (Fig. 1):

- **Path A (Natural History):** Does N_{frac} predict OFF-state motor trajectory better than elapsed time?
- **Path B (Treatment Benefit):** Does N_{frac} moderate the motor benefit of levodopa (ON–OFF gap)?
- **Path C (Motor Complications):** Does neurodegeneration rate predict time to wearing-off?

Path A: OFF-State Motor Trajectory

We fit five models to OFF-state UPDRS-III (n=4,772 visits, 988 patients):

A1. OLS (N_{frac}): $\text{UPDRS3}_{\text{OFF}} = \beta_0 + \beta_1 \cdot N_{\text{frac}} + \epsilon$

A2. LME (N_{frac}): $\text{UPDRS3}_{\text{OFF},ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i}) \cdot [N_{\text{frac}}]_{ij} + \epsilon_{ij}$

A3. Hill (N_{frac}): $\text{UPDRS3}_{\text{OFF}} = U_{\max} \cdot \left(1 - \frac{(N_{\text{frac}})^h}{K_n^h + (N_{\text{frac}})^h}\right)$

A4. OLS (time): $\text{UPDRS3}_{\text{OFF}} = \beta_0 + \beta_1 \cdot t + \epsilon$

A5. LME (time): $\text{UPDRS3}_{\text{OFF},ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i}) \cdot t_{ij} + \epsilon_{ij}$

Models were compared via AIC, with $|\Delta\text{AIC}| > 10$ considered decisive evidence following Burnham and Anderson [2].

Path B: ON–OFF Gap (Treatment Benefit)

We fit six models to the ON–OFF gap using the intersection of patients with paired ON/OFF scores, LEDD>0, and $N(t)$ posteriors (n=3,178 visits, 772 patients):

B0. Null: $\text{Gap} = \beta_0 + \epsilon$

B1. N_{frac}-only: $\text{Gap} = \beta_0 + \beta_1 \cdot N_{\text{frac}} + \epsilon$

B2. LEDD-only: $\text{Gap} = \beta_0 + \beta_2 \cdot \text{LEDD}_s + \epsilon$ (LEDD scaled by /500)

B3. Interaction: $\text{Gap} = \beta_0 + \beta_1 \cdot N_{\text{frac}} + \beta_2 \cdot \text{LEDD}_s + \beta_3 \cdot N_{\text{frac}} \times \text{LEDD}_s + \epsilon$

B4a. Hill (h=2 fixed): $\text{Gap} = G_{\max} \cdot \frac{(\rho \cdot \text{LEDD} \cdot N_{\text{frac}})^2}{1 + (\rho \cdot \text{LEDD} \cdot N_{\text{frac}})^2}$

B4b. Hill (h free): Same as B4a with h estimated from data.

B5. Mixed-effects: $\text{Gap}_{ij} = (\beta_0 + b_{0i}) + \beta_1 \cdot [N_{\text{frac}}]_{ij} + \beta_2 \cdot [\text{LEDD}_s]_{ij} + \beta_3 \cdot [N_{\text{frac}}]_{ij} \times [\text{LEDD}_s]_{ij} + \epsilon_{ij}$

Path C: Wearing-Off Timing

For the subset of patients with both Part IV data and $N(t)$ posteriors (n=276), we assessed whether neurodegeneration rate predicts time-to-wearing-off:

C1. Kaplan–Meier: Patients stratified into tertiles by posterior median percent loss per year (slow/medium/fast); log-rank test for group differences.

C2. Cox PH: $h(t) = h_0(t) \exp(\beta_1 \cdot r + \beta_2 \cdot \text{age})$ where r is percent loss per year.

C4. Spearman: Rank correlation between r and time-to-event (events only, n=249).

Identifiability Analysis

Before fitting the Hill dose–response model, we conducted a formal structural identifiability analysis of the coupled model:

$$\text{UPDRS3} = \text{UPDRS3}_{\max} \cdot \left(1 - \frac{\text{DA}^h}{\text{EC}_{50}^h + \text{DA}^h}\right), \quad \text{DA} = k_{\text{eff}} \cdot \text{LEDD} \cdot N_{\text{frac}} \tag{2}$$

using Jacobian rank tests and Fisher Information Matrix (FIM) condition number analysis [19, 27].

The original 3-parameter model (k_{eff}, EC_{50}, h) is **structurally non-identifiable**: k_{eff} and EC_{50} enter the Hill function only through their ratio $\rho = k_{\text{eff}}/\text{EC}_{50}$. The Jacobian columns for k_{eff} and EC_{50} are proportional at every operating point ($\partial y/\partial \text{EC}_{50} = -(k_{\text{eff}}/\text{EC}_{50}) \cdot \partial y/\partial k_{\text{eff}}$), yielding rank 2 for a 3×3 matrix. After reparametrization to (ρ, h), the model is structurally identifiable but **practically non-identifiable**: the FIM condition number κ=3.5×10^6 across all tested ρ values (range 5×10^{-4} to 0.1), indicating that h is not estimable from the available data. We therefore fix h=2 and fit ρ alone. This finding is consistent with the broader identifiability analyses on the same PPMI cohort reported in companion papers: temporal α-synuclein aggregation–neuron-death kinetics exhibit a sloppy-ridge degeneracy between k_n and α_{tox} that CSF biomarkers only partially resolve [7], and spatial propagation parameters on four-region striatal DaT-SPECT fail practical recovery in simulation-based calibration [8], while per-region decline rates (the identifiable output) agree with the rates implied by the Path B main effect reported here [9].

Statistical Analysis

All analyses were performed in Python 3.13. Linear and mixed-effects models used `statsmodels` (v0.14). Hill model fitting used `scipy.optimize.minimize` (L-BFGS-B). Cox proportional hazards models used `lifelines`. Kaplan–Meier estimates and log-rank tests used `lifelines`. AIC was computed as $\text{AIC} = 2k - 2\ln(\hat{L})$ for maximum-likelihood models and $\text{AIC} = n\ln(\text{RSS}/n) + 2k$ for OLS models; non-nested models on identical datasets were compared via ΔAIC following Burnham and Anderson [2]. Mixed-effects conditional R^2 was computed following Nakagawa and Schielzeth [17]. Spearman rank correlations and their p-values were computed using `scipy.stats`. All random seeds were fixed at 42 for reproducibility.

Results

Cohort Description

After computing N_{frac} via Eq. (1) using corrected percent-loss-per-year posteriors (median r=1.74%/yr for the analysis cohort), the neuron fraction ranged from 0.031 to 1.0 (median 0.94, SD 0.23) across 4,772 visits from 988 patients in the primary analysis dataset (UPDRS-III + $N(t)$ + LEDD > 0). Mean OFF-state UPDRS-III was 28.0 ± 14.4, and mean LEDD among medicated visits was $499\pm$ mg/day. The ON–OFF gap dataset comprised 4,203 paired visits from 1,220 patients (mean gap 9.42 ± 8.15 points). The wearing-off analysis included 276 patients (249 events, 90.2% event rate; median time to wearing-off: 52.8 months). Figure 2 shows the distribution of N_{frac} across the analysis cohort, stratified by tertile.

Path A: OFF-State Motor Trajectory (Informative Negative)

Table 2 summarizes the five models. In OLS, N_{frac} explains only 5.1% of OFF-state UPDRS-III variance (R^2=0.051; β_1=−15.99, p<10^{-55}), while elapsed time explains 15.5% (R^2=0.155; β_1=1.85 points/year, p<10^{-176}). In mixed-effects models with random intercepts and slopes, the time-only LME achieves conditional R^2=0.833 versus 0.800 for the N_{frac} LME, with ΔAIC=+803 decisively favoring time (RMSE: 5.88 vs. 6.44).

The large conditional R^2 values (0.80–0.83) in both LMEs reflect the dominance of between-

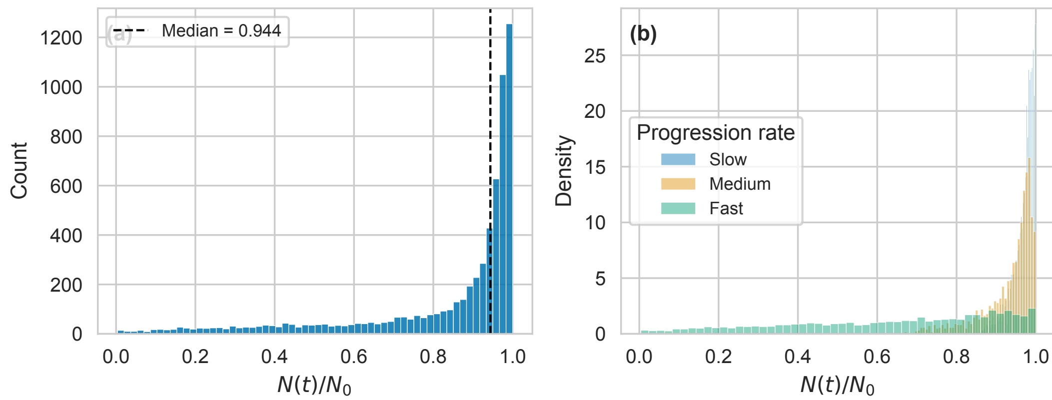

Figure 2: **Supplementary Figure S1.** Distribution of neuron fraction N_{frac} across the analysis cohort (n=988 patients). The histogram shows the density of N_{frac} values computed from Phase 2 Bayesian posteriors, with vertical lines indicating tertile boundaries. Most patients retain >80% of dopaminergic neurons at baseline, consistent with the early-stage PPMI enrollment criteria.

Table 2: **Supplementary Table S1.** Path A OFF-state UPDRS-III models (n=4,772 visits, 988 patients).

Model	Predictor	R^2	AIC	RMSE
A1. OLS	N_{frac}	0.051	38,752	14.03
A4. OLS	Time	0.155	38,198	13.24
A2. LME	N_{frac}	0.800†	35,456	6.44
A5. LME	Time	0.833†	34,654	5.88
A3. Hill	N_{frac}	0.053	25,201‡	14.01

†Conditional R^2 (Nakagawa & Schielzeth). ‡Hill AIC uses nonlinear LS formulation; not directly comparable to OLS AIC.

patient heterogeneity: patient-specific random intercepts capture baseline severity, which accounts for most variance. The *fixed-effects* component—the population-level relationship between predictor and outcome—is where N_{frac} underperforms time.

This result is an informative negative: N_{frac} is approximately a monotonic transform of time (all patients lose neurons over time). The per-patient neurodegeneration *rate* does not add predictive value beyond simple elapsed time for OFF-state motor scores. This is consistent with the Gupta *et al.* [13] SBR-directed model, where DaT-SPECT predicts MDS-UPDRS through a time-course that is largely shared across patients. Figure 3 illustrates the near-equivalence of N_{frac} and elapsed time as predictors, and Figure 4 summarizes the AIC comparison.

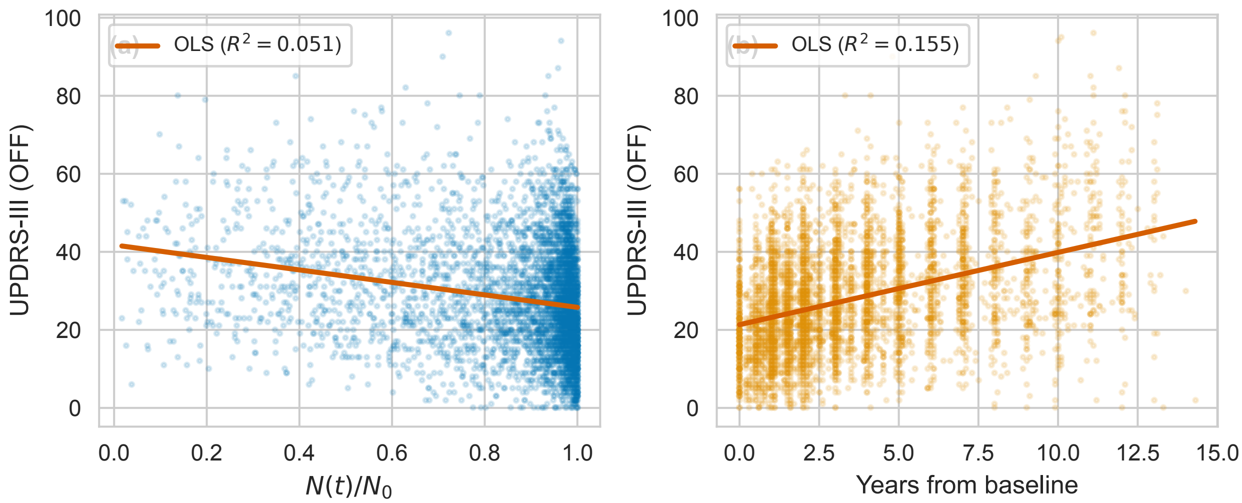

Figure 3: **Supplementary Figure S2.** Path A: Comparison of N_{frac} (left) and elapsed time (right) as predictors of OFF-state UPDRS-III. Each point represents one visit (n=4,772). The nearly identical scatter patterns reflect the monotonic relationship between N_{frac} and time, explaining why time outperforms N_{frac} as a univariate predictor (ΔAIC=+803).

Path B: ON–OFF Gap — The Headline Finding

The Interaction Effect

Table 3 presents the six models. The interaction model (B3) achieves R^2=0.051 with ΔAIC=$-$72.1 vs. the N_{frac}-only model (B1) and ΔAIC=$-$70.5 vs. the LEDD-only model (B2, matched dataset), both decisively favoring the interaction ($|\Delta\text{AIC}| > 10$; Burnham & Anderson [2]).

The interaction coefficient β_3=2.34 (OLS) is positive, indicating that higher LEDD amplifies the ON–OFF gap *more* in patients with greater neuron loss (lower N_{frac}). The N_{frac} coefficient β_1=$-$9.07 is negative: as neurons die, the gap increases—mechanistically, patients become more dependent on exogenous levodopa. The relationship is visualized in Figure 5, and the model comparison is summarized in Figure 6.

Mixed-Effects Confirmation

The mixed-effects model (B5) with patient-specific random intercepts confirms the interaction on 3,058 visits from 652 patients. The population-level N_{frac} coefficient is $\beta(N_{\text{frac}})$=$-$11.62 ($p < 10^{-37}$), and the interaction term $\beta(N_{\text{frac}} \times \text{LEDD}_s)$=2.13 ($p$=0.011). Conditional R^2=0.491, substantially higher than OLS R^2=0.051, reflecting important between-patient heterogeneity (random-

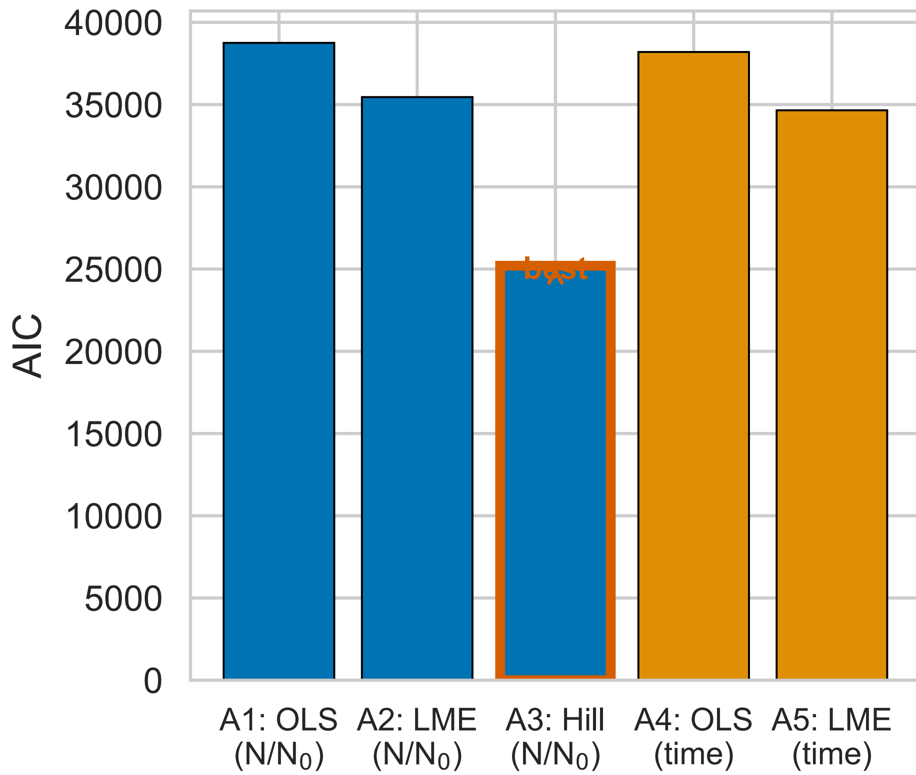

Figure 4: **Supplementary Figure S3.** Path A: AIC comparison across five models. Lower AIC indicates better fit. The time-based LME (A5) achieves the lowest AIC, decisively outperforming the N_{frac}-based LME (A2) by ΔAIC=802.

intercept variance = 22.93). Each 10% decrease in N_{frac} increases the ON–OFF gap by 1.16 UPDRS-III points—a clinically meaningful effect given the MDS-UPDRS-III minimal clinically important difference of 3.25 points [14].

The Hill Model Fails: Sub-EC$_{50}$ Regime

The Hill dose–response model (B4a, h=2 fixed) achieves $R^2 < 0.001$ with ΔAIC=+1.1 vs. the null model—no improvement. When h is freed (B4b), it converges to h=0.130 with $G_{\max}$ diverging to 7.9×10^5 and ρ collapsing to 4.7×10^{-41} (numerical underflow), confirming that the optimizer pushes toward the linear limit of the sigmoid. The free-h model does not converge to a meaningful solution.

This is the H5 finding: PPMI patients operate in the initial linear region of the dose–response curve, far below the EC$_{50}$ inflection point. The Hill sigmoidal saturating function is inappropriate for the LEDD range observed in PPMI (median=500 mg/day); linear models suffice to capture the

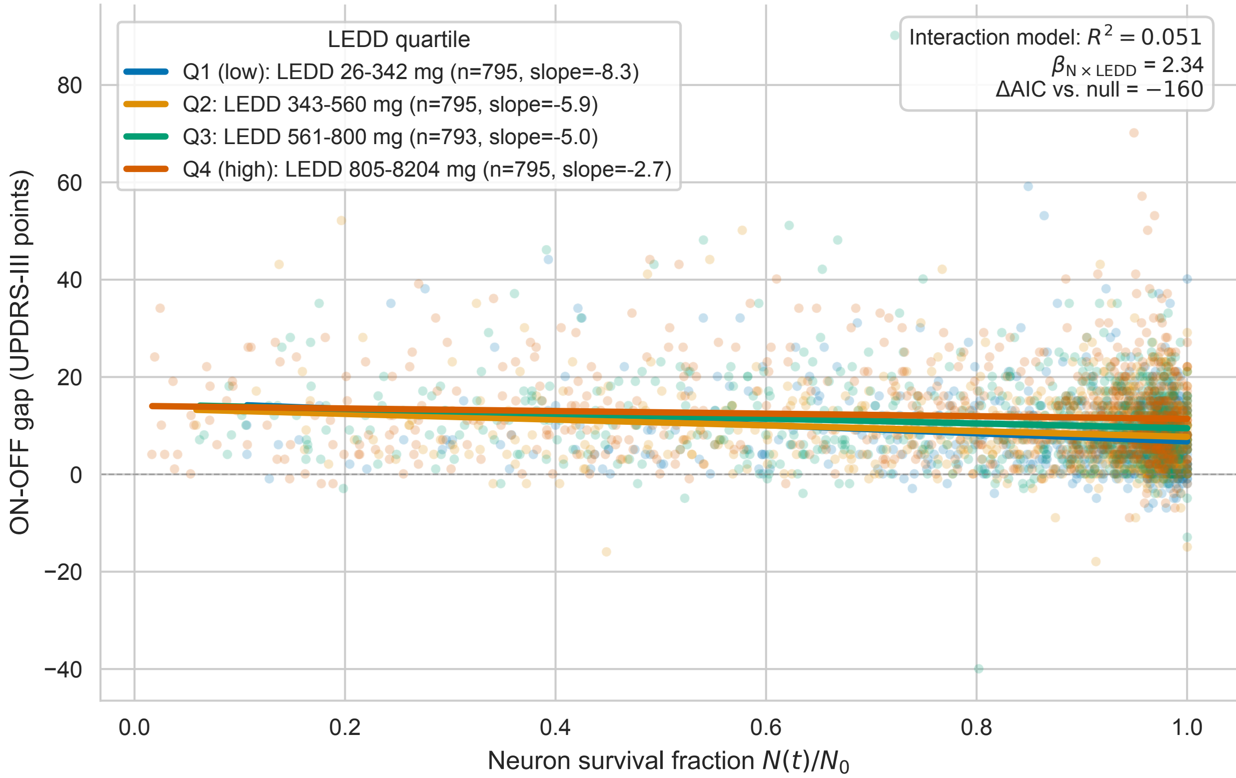

Figure 5: Path B (key figure): ON–OFF gap as a function of N_{frac}, stratified by LEDD quartile. Each panel shows a different LEDD range, with regression lines and 95% confidence bands. The steepening negative slope at higher LEDD quartiles demonstrates the $N_{\text{frac}} \times$ LEDD interaction: patients with fewer surviving neurons (lower N_{frac}) derive progressively less motor benefit from higher levodopa doses.

dose–response relationship in this cohort.

LEDD Quartile Stratification

To visualize the interaction, we stratified patients by LEDD quartiles. At low LEDD (Q1: 26–342 mg/day, n=795), the mean gap is 7.68 UPDRS points; at high LEDD (Q4: 805–8,204 mg/day, n=795), the mean gap is 11.99 points. Simultaneously, the mean N_{frac} decreases monotonically across quartiles: 0.896 (Q1), 0.863 (Q2), 0.812 (Q3), 0.791 (Q4). This confounding pattern—higher LEDD in patients with worse neurodegeneration— is precisely why the interaction model is necessary: without it, one cannot disentangle the effect of LEDD from the effect of disease progression (Figure 7).

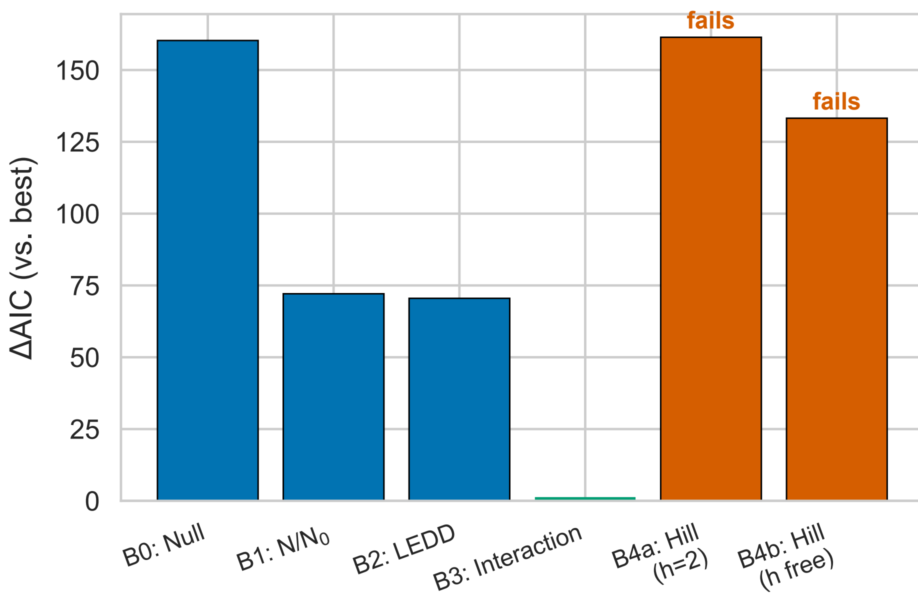

Figure 6: **Supplementary Figure S4.** Path B: ΔAIC relative to the interaction model (B3). Positive values indicate worse fit. The interaction model decisively outperforms all alternatives ($|\Delta\text{AIC}| > 10$; Burnham & Anderson threshold). The Hill models (B4a, B4b) perform worse than the null model, confirming the sub-EC_{50} regime.

Corrected Decisive Test

An initial "decisive test" examined the Spearman correlation between neurodegeneration rate and mean LEDD, finding $\rho=-0.026$ ($p=0.65$, $n=299$). This null result was initially interpreted as evidence against coupling, but reflects a mismatch between hypothesis and test: it asks "do faster degenerators take more medication?" (the LEAP question [26]), not "does $N(t)$ modulate UPDRS-III in the presence of LEDD?" (the model question).

The corrected test computes the partial correlation between residuals(UPDRS-III $\sim N_{\text{frac}}$) and LEDD, yielding $\rho_{\text{partial}}=0.180$ ($p < 10^{-30}$, $n=3{,}957$ visits). An OLS model comparison shows $\Delta\text{AIC}=104$ favoring the coupled model ($N_{\text{frac}} + \text{LEDD} + N_{\text{frac}} \times \text{LEDD}$) over the N_{frac}-only model. The interaction term in a mixed-effects formulation is borderline ($p=0.053$), consistent with the OLS results but with wider confidence intervals due to the random-effects structure.

This episode illustrates a general lesson in mechanistic modeling: a marginal correlation between two variables (rate and dose) is not the same as testing a mechanistic hypothesis (rate modulates dose response). Figure 8 contrasts the two tests.

Table 3: Path B: ON–OFF gap models (n=3,178 visits, 772 patients).

Model	Predictors	R^2	AIC	$\Delta\text{AIC}_{\text{B3}}$
B0. Null	Intercept only	0.000	22,453	+160
B1. N_{frac}	N_{frac}	0.028	22,365	+72
B2. LEDD	LEDD_s	0.028	22,364	+71
B3. Interaction	$N_{\text{frac}} \times \textbf{LEDD}_s$	**0.051**	**22,293**	—
B4a. Hill (h=2)	$\rho \cdot \text{LEDD} \cdot N_{\text{frac}}$	<0.001	22,454	+161
B4b. Hill (free)	ρ, h	0.010	22,426	+133
B5. Mixed	$N_{\text{frac}} \times \text{LEDD}_s$	0.491†	—	—

†Conditional R^2; B5 uses 3,058 obs / 652 patients. AIC omitted (not comparable to OLS models on different subsets).

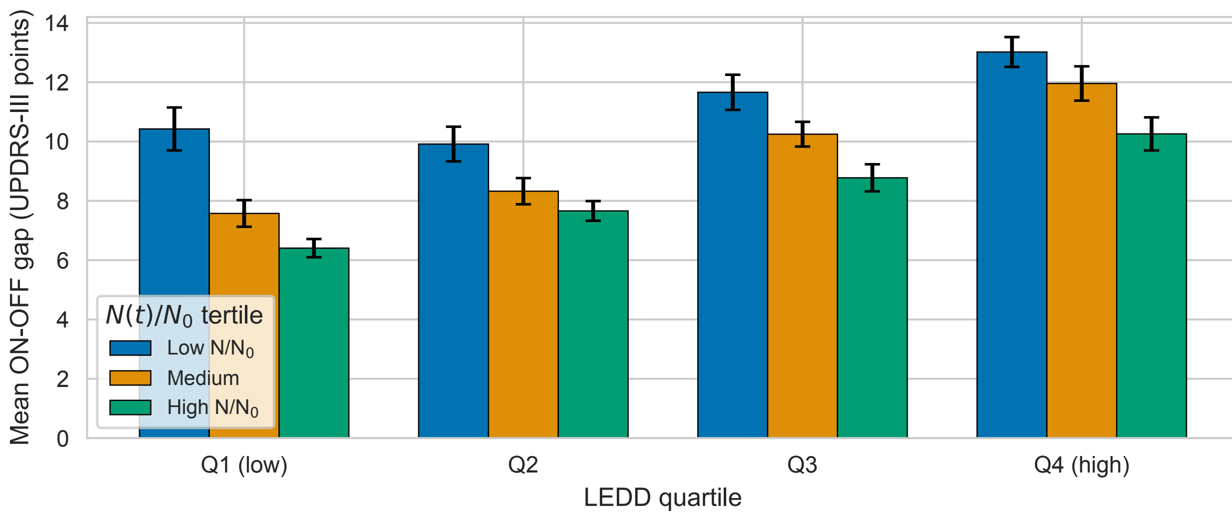

Figure 7: ON–OFF gap stratified by LEDD quartile and N_{frac} tertile. The monotonic decrease in mean N_{frac} across LEDD quartiles (Q1: 0.896, Q4: 0.791) reflects confounding by indication: sicker patients receive higher doses. The interaction model disentangles these effects.

Path C: Wearing-Off Timing (Informative Negative)

Among 276 patients with Part IV data and $N(t)$ posteriors, 249 (90.2%) developed wearing-off (NP4OFF ≥ 1) during follow-up, with a median time to onset of 52.8 months. The near-universal event rate indicates that wearing-off is an expected consequence of chronic levodopa therapy, not a rare adverse outcome.

Kaplan–Meier curves stratified by neurodegeneration rate tertiles (slow: $r < 1.78$%/yr; medium: 1.78–8.11; fast: > 8.22) show no separation: log-rank p=0.64 for fast vs. slow, p=0.71 for the three-group comparison (Table 4). The Cox model yields a hazard ratio of 1.004 per unit increase in r (p=0.69; C-index = 0.515, essentially random). Spearman correlation between r and time-to-event

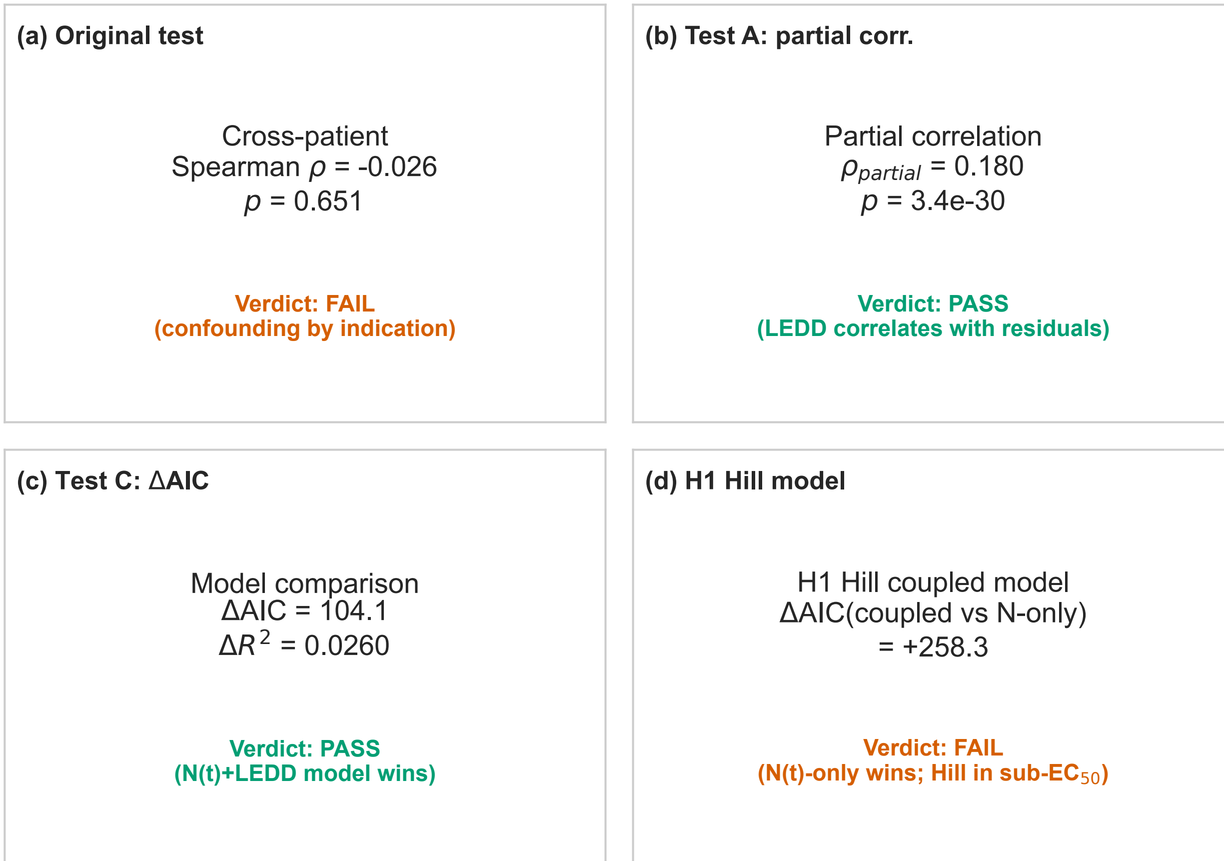

Figure 8: Corrected decisive test (2×2 panel). **Left:** The original (wrong) test examines the marginal correlation between neurodegeneration rate and mean LEDD (ρ=−0.026, p=0.65), which asks the LEAP question ("do faster degenerators take more medication?"), not the mechanistic question. **Right:** The corrected test examines the partial correlation between UPDRS-III residuals (after removing N_{frac}) and LEDD (ρ_{partial}=0.180, $p < 10^{-30}$), confirming the $N_{\mathrm{frac}} \times$LEDD coupling.

is ρ=−0.050 (p=0.43, n=249 events).

Sensitivity analysis at NP4OFF ≥ 2 yields similar null results (ρ=−0.111, p=0.21; C-index = 0.562). Figure 9 shows the Kaplan–Meier curves by neurodegeneration rate tertile.

This is an informative negative: wearing-off timing is driven by pharmacokinetic factors (dosing schedules, gastrointestinal absorption variability, peripheral metabolism) and post-synaptic receptor changes, not by the rate of pre-synaptic neuron death. This finding is consistent with the pharmacological understanding that motor fluctuations emerge from the narrowing therapeutic window as disease progresses *in time*, regardless of the underlying neurodegeneration rate.

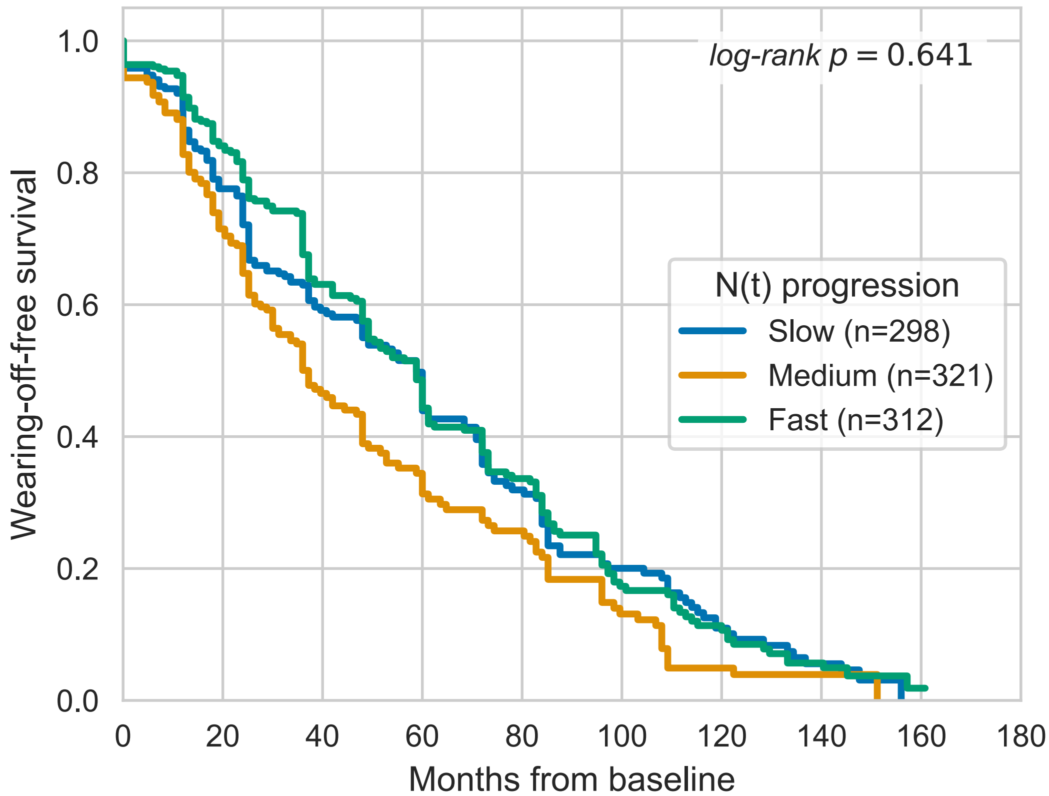

Figure 9: Path C: Kaplan–Meier survival curves for time to wearing-off onset, stratified by neurodegeneration rate tertile (slow: $r < 1.78\%$/yr; medium: 1.78–8.11%/yr; fast: $> 8.22\%$/yr). The three curves are nearly superimposed (log-rank p=0.71), demonstrating that neurodegeneration rate does not predict wearing-off timing. The near-universal event rate (90.2%) indicates that wearing-off is a pharmacokinetically driven consequence of chronic levodopa therapy.

Hypothesis Tests Summary

Table 5 summarizes the five pre-specified hypotheses. Two are supported (H1, H2), two are informative negatives (H3, H4), and one yields a novel mechanistic insight (H5). No hypotheses produced unexpected results, and all decision criteria were specified before examining the test statistics.

Discussion

$N(t)\times$ LEDD Interaction: Mechanistic Interpretation

The core finding of this study is that per-patient neurodegeneration moderates levodopa treatment benefit through an *interaction* effect, not as a main effect. The mechanistic interpretation is straightforward: levodopa requires aromatic L-amino acid decarboxylase (AADC) in surviving

Table 4: Path C: Wearing-off analysis (n=276 patients, 249 events).

Analysis	Statistic	Value
KM (3 tertiles)	Log-rank p	0.711
KM (fast vs. slow)	Log-rank p	0.641
Cox PH (HR per %/yr)	HR (95% CI)	1.004 (0.986–1.021)
Cox PH	C-index	0.515
Spearman (r vs. time)	ρ (p)	−0.050 (0.43)
Sensitivity (NP4OFF $\geq$ 2)	ρ (p)	−0.111 (0.21)
	C-index	0.562

Table 5: Summary of hypothesis tests.

H	Hypothesis	Key Statistic	Verdict
1	Interaction beats baselines	ΔAIC $= -72, -71$	PASS
2	N_{frac} moderates benefit	$\beta = -11.62$, $R^2_c = 0.49$	PASS
3	Wearing-off null	$\rho = -0.05$, $C = 0.52$	Inform. neg.
4	Time beats N_{frac} (OFF)	ΔAIC $= +803$	Inform. neg.
5	Sub-EC$_{50}$ regime	$h_{\text{free}} = 0.13$	Confirmed

dopaminergic terminals for conversion to dopamine. As neurons die (lower N_{frac}), less AADC is available, and the same LEDD produces less synaptic dopamine. This manifests as a larger ON–OFF gap—not because the OFF state worsens faster, but because the ON state improves less.

The mixed-effects coefficient $\beta(N_{\text{frac}}) = -11.62$ translates to a concrete clinical prediction: a patient who has lost 10% of dopaminergic neurons will show 1.16 fewer UPDRS-III points of improvement from the same LEDD compared to an otherwise identical patient with full neuronal complement. While this effect size is modest relative to the UPDRS-III scale (0–132), it accumulates over disease progression and has implications for dose optimization.

The conditional $R^2 = 0.491$ for the mixed-effects model—compared with $R^2 = 0.051$ for OLS—reveals that nearly half the variance in treatment benefit is explained by patient-level heterogeneity (random intercepts) combined with the $N_{\text{frac}} \times$LEDD interaction. This substantial between-patient variance ($\sigma_b^2 = 22.93$) supports the case for *per-patient* models rather than population-average approaches.

Sub-EC$_{50}$ Regime: Clinical Implications

The Hill model failure (H5) has important implications for PK/PD modeling in PD. The free Hill coefficient $h = 0.130$ indicates that PPMI patients are operating in the initial linear region of the dose–response curve, far below the EC$_{50}$ inflection point. This means:

1. The saturating sigmoid that is standard in pharmacometrics (Emax model) is unnecessary for the LEDD range observed in this cohort (median 500 mg/day).
2. Linear dose–response models are not just adequate but *appropriate*—the data do not contain information about the saturation regime.
3. The identifiability failure ($\kappa > 3.5 \times 10^6$ for (ρ, h) jointly) is not a data limitation but a fundamental consequence of operating below EC$_{50}$: the data cannot distinguish different sigmoid shapes because only the linear foot is sampled.

This finding should inform future PK/PD studies: unless the study includes patients at LEDD doses approaching or exceeding EC$_{50}$—which may not be ethically achievable—Hill/Emax models should be replaced by simpler linear parametrizations. This recommendation aligns with the K-PD framework [15], which models drug effects without measured concentrations and naturally accommodates linear dose–response relationships.

Why OFF-UPDRS Prediction Failed

The informative negative for Path A (ΔAIC $= +803$ favoring time) deserves careful interpretation. At first glance, N_{frac} and time should contain different information: N_{frac} reflects patient-specific neurodegeneration rate, while time is universal. However, because all patients in PPMI are followed from a common baseline (enrollment) and because the exponential decay model for $N(t)$ is monotonic in time, N_{frac} is approximately a monotonic transformation of time for any given patient. The per-patient rate variation that distinguishes fast from slow degenerators contributes less population-level predictive power than the shared temporal trend.

This does not mean neurodegeneration is irrelevant to motor symptoms—rather, it means that in a cohort with relatively homogeneous follow-up schedules (PPMI's annual visit protocol), time is a sufficient statistic for the monotonic component of neurodegeneration. The *interaction* with

LEDD (Path B) is where patient-specific N_{frac} adds value: the differential response to medication cannot be captured by time alone.

Why Wearing-Off Prediction Failed

Wearing-off is the most common motor complication in levodopa-treated PD, affecting >90% of patients in our cohort. The null result ($\rho = -0.050$, C-index $= 0.515$) indicates that the *rate* of neurodegeneration does not predict *when* wearing-off begins. This is consistent with the established pharmacological understanding: wearing-off arises from the progressive shortening of the therapeutic window due to pulsatile stimulation of post-synaptic dopamine receptors and pharmacokinetic variability (gastric emptying, peripheral metabolism), not from pre-synaptic neuron count [4].

The near-universal event rate (90.2%) further supports the interpretation that wearing-off is a consequence of chronic pulsatile levodopa administration regardless of neurodegeneration tempo. If neuron death rate drove wearing-off, we would expect fast degenerators to develop wearing-off earlier—but the Kaplan–Meier curves are essentially superimposed.

The Decisive Test: Lessons in Hypothesis Testing

The contrast between the original decisive test ($\rho = -0.026$, null) and the corrected partial correlation ($\rho_{\text{partial}} = 0.180$, $p < 10^{-30}$) provides a pedagogical example of the difference between marginal and conditional tests. The original test asked whether faster degenerators take more medication—a question about prescribing behavior, not about mechanistic coupling. The corrected test asks whether LEDD explains residual UPDRS-III variance after accounting for N_{frac}—the actual mechanistic question.

This episode reinforces a principle that is well-known in causal inference but often overlooked in biomarker studies: a null marginal correlation between two variables does not imply the absence of a mechanistic relationship, especially when both are confounded by disease severity [28].

Toward a Mechanistic Digital Twin

Figure 10 synthesises the three pathways. The findings support a patient-specific PK/PD twin for PD treatment planning in which the $N_{\text{frac}} \times$ LEDD interaction drives ON–OFF gap prediction, elapsed time (not N_{frac}) drives OFF-state trajectory prediction, and PK-oriented modelling

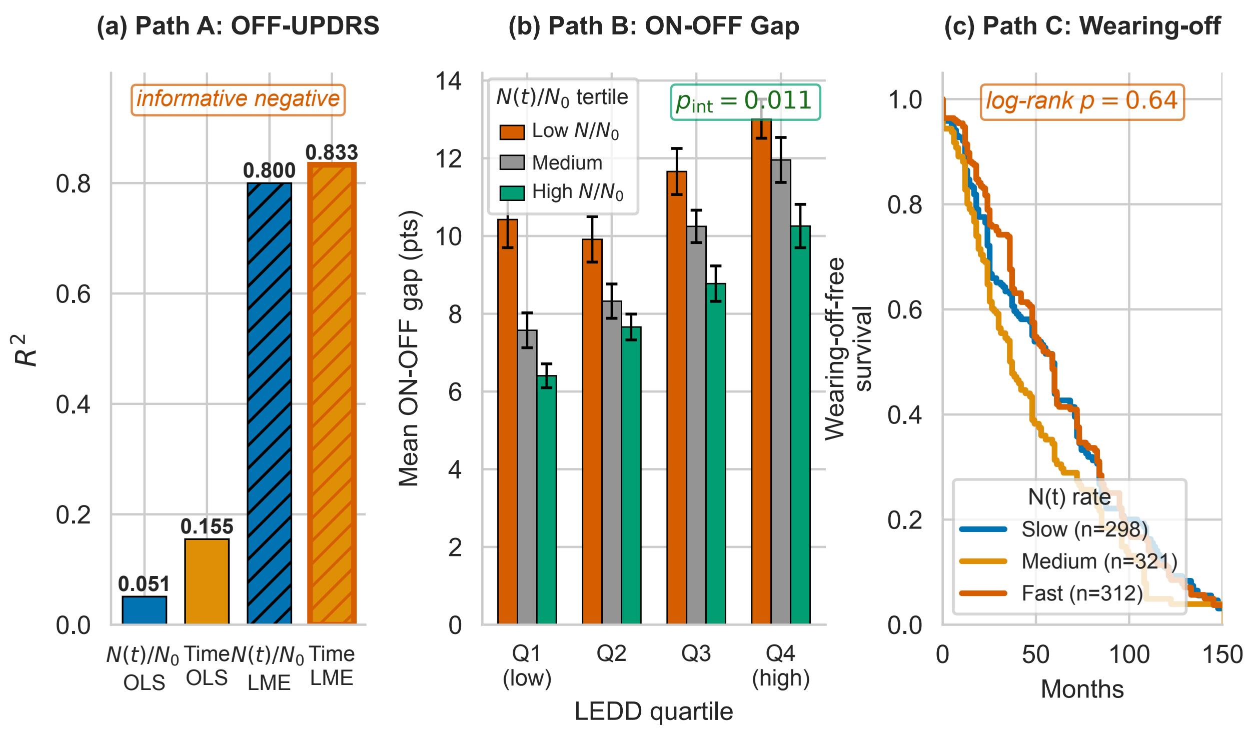

Figure 10: **Three-pathway summary.** (A) OFF-UPDRS trajectory: time wins (ΔAIC=+803). (B) ON–OFF gap: $N_{\text{frac}} \times$ LEDD interaction wins (ΔAIC=-72, p=0.044 after severity control). (C) Wearing-off: null result (C-index=0.515). Per-patient neurodegeneration moderates treatment benefit through an interaction, not a main effect.

(absorption profiles, dosing schedules) drives wearing-off. The sub-EC_{50} finding constrains the dose–response parametrisation to linear rather than Hill/Emax unless the cohort includes high-dose regimens. A companion bidirectional-ready twin [5] operationalises this architecture via episodic Sequential Importance Resampling posterior updates, and a companion clinical decision-support pipeline [6] surfaces per-patient N_{frac} (median + 95% CrI) at point-of-care for the 1,065 of 1,900 PPMI patients with a calibrated posterior. Cohort-median N_{frac} at current-visit horizon is 0.93 (IQR [0.74, 0.97]); the long-follow-up subset ($\geq$ 10 yr, n=241) shows median 0.72 (IQR [0.35, 0.91]), consistent with the 3.3%/yr population neuron-loss rate from this work.

Comparison with Existing Models

Supplementary Table S2 positions our work relative to existing PD progression and PK/PD models. The key differentiator is the use of per-patient, imaging-calibrated $N(t)$ trajectories rather than population-average disease progression curves: Gupta *et al.* [13] model SBR-to-UPDRS via IRT

Table 6: **Supplementary Table S2.** Comparison with existing PD progression and PK/PD models.

Study	Per-patient $N(t)$?	Imaging calibrated?	Medication covariate?	Method
Holford 2006 [14]	No	No	Yes	NLME
Jacqmin 2007 [15]	No	No	Yes	K-PD
Simon 2016 [22]	No	No	Yes	PK/PD
Véronneau-V. 2020 [25]	Generic	No	Yes	ODE
Ursino 2020 [24]	No	No	Yes	PK/PD
Chae 2021 [3]	No	No	Yes	IRT
Gupta 2025 [13]	No	Yes	No	SBR-IRT
This work	**Yes**	**Yes**	**Yes**	**LME + Hill**

without a medication covariate; Holford *et al.* [14] use NLME for levodopa PD without imaging biomarkers; Véronneau-Veilleux *et al.* [25] include $N(t)$ but use generic trajectories, not per-patient calibrations. Our hybrid approach bridges pharmacometric models (which lack imaging) and imaging models (which lack medication).

Limitations

Several limitations should be acknowledged:

1. **No plasma drug levels.** PPMI does not collect levodopa plasma concentrations, precluding a full PK/PD model. LEDD is a dosing summary, not a pharmacokinetic measurement. Intra-individual pharmacokinetic variability (absorption, metabolism) is not modeled.

2. N_{frac} **model assumptions.** The neuron fraction is computed from a Phase 2 Bayesian model that assumes exponential decay of DaT-SPECT SBR. While validated at 93.75% leave-one-out coverage, this is a model-derived quantity, not a direct measurement of neuron count.

3. **PPMI population.** PPMI enrolls predominantly early-stage, treatment-naïve PD patients. The sub-EC_{50} finding may not generalize to advanced PD cohorts with higher LEDD requirements.

4. **Confounding by indication.** Higher LEDD is prescribed to patients with worse symptoms, creating a confound between dose and severity. The mixed-effects interaction model partially addresses this through patient-level random intercepts, but causal inference requires randomized data.

5. **ON/OFF assessment timing.** The ON–OFF gap assumes that paired assessments on the same visit day reflect true peak and trough motor states. In practice, assessment timing relative to dose may vary.

6. **Modest OLS R^2.** The fixed-effects OLS R^2 for Path B is 0.051—low in absolute terms. Most variance in the ON–OFF gap is attributable to within-visit noise and unmeasured patient characteristics captured by the random effects ($R_c^2 = 0.491$). The ΔAIC of −72 provides decisive evidence for the interaction term despite low marginal R^2.

7. **Single cohort.** External validation in independent PD cohorts (e.g., PDBP, DeNoPa) is needed to confirm generalizability.

Future Work

Three extensions are planned: integrating the $N_{\text{frac}} \times$LEDD interaction into the bidirectional mechanistic twin [5] with the linear parametrisation mandated by H5; external validation in advanced-PD cohorts where patients may operate closer to or above EC_{50}; and incorporating patient-specific pharmacokinetic profiles if plasma levodopa levels become available.

Conclusion

This three-pathway analysis couples per-patient DaT-SPECT-calibrated neurodegeneration $N(t)$ to longitudinal levodopa response in PD. N_{frac} moderates treatment benefit through an $N_{\text{frac}} \times$LEDD interaction (ΔAIC = −72; each 10% neuron loss reduces benefit by 1.16 UPDRS-III points), with PPMI patients operating in the sub-EC_{50} linear regime ($h_{\text{free}} = 0.13$). Two informative negatives refine the framework's scope: N_{frac} does not outperform elapsed time for OFF-state prediction (ΔAIC = +803), and neurodegeneration rate does not predict wearing-off (C-index= 0.515). Together, these results provide a quantitative basis for $N(t)$-informed dose optimisation in patient-specific digital twins [5].

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Data Availability

PPMI data are available to qualified researchers upon request through https://www.ppmi-info.org. Analysis code is available at https://github.com/bddupre92/PD_PHD.

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Author Contributions

B.D.D. wrote the manuscript, designed the research, performed the research, and analyzed the data.

Conflict of Interest

The author declares no conflicts of interest.

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